

T2J-Bench: A Bug-Fix Benchmark for PyTorch-to-JAX Code Translation

Important research questions:

Prerequisite:

- Dataset of bugs and fixed bugs for Torch-to-JAX code translation.

Research question 1: JAX bug fixing accuracy.

Research question 2: Code Translation accuracy.

Tasks

Task 1: Revising the dataset of Torch-to-JAX buggy datasets.

- <https://huggingface.co/spaces/bigcode/bigcode-models-leaderboard>
- Regenerating translated code with recent open LLMs: GRM

Step 1: Selecting PyTorch dataset.

Step 2: getting JAX translated code from openLLMs: qwen, mistral etc. GLM-5.2 ([z.ai](#))

Evaluate (**weak**) open LLM (junior) on torch 2 jax, determine good ones? (output faithful codes, accept trivial errors).

- Tasks are sufficiently challenging. (Torch Leet (code interview, **LLM**), **KernelBench**)
 - Code snippets (short context)
 - Not repo - multiple files? (not long context)
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- Consider < **10B** as weak. (depending on the complexity of tasks).
- Figure: `LLM size` vs **quality of code translation**
 - Caveat: LLM size is not only factor, but we can't control the knowledge / training data
 - 5 -10 models
 - Temperature = 0.7
- How to know if it's **trivial** code (should filter-out) , only keep **compilable** code?
 - Trivial code (ask Claude to build verifier)
- Second direction: tensorflow2torch (big system), torch2tensorflow (on-device).
 - RL - framework (torch) ⇒ jax?
 - Framework (tensorflow) ⇒ torch? - ask Claude ⇒ most popular machine learning framework (focus LLM)
- *Triton - OpenAI backend*
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- **What openLLMs?**

- Prompt for code translation

Step 3: Evaluating percentage of JAX translated code:

- Failed for compilation.
- Failed for correctness.

Task 2: Doing bug fixing by agent:

- Agent is scalable, human is not
 - Claude, GPT-5.4, GLM-5.2, DeepSeek as strong LLM (senior)
- Writing prompt for doing the fixing.
 - Humans write code verifiers.
 - Use weak LLM (Task 1) (least trivial code but many compilable code)
- Junior fixes in n steps and says it's correct / terminate if failed after n steps. The senior jumps in and fixes it.
 - Buggy jax (incorrect jax, error messages)
 - **after-fixing jax** (after fixing above error message, not final version) is output accepted by senior LLM
- Our **real system**:
 - Code-translation LLM
 - Verifier
 - compiler

Task 3: Code translation:

- Given the input as
 - **PyTorch code**
 - JAX prelim translated code (+ error messages) ⇒
- output: final JAX code
- **Step 1:** preparing the fine-tuning process on sample data.